

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problem

Description: You are given an array of integers `nums` and an integer `k`.

An *inversion pair with a threshold x* is defined as a pair of indices (i, j) such that:

- (i) $i < j$,
- (ii) `nums[i] > nums[j]`, and
- (iii) the difference between the two numbers is at most x , i.e.

$$\text{nums}[i] - \text{nums}[j] \leq x.$$

Your task is to determine the minimum integer `min_threshold` such that there are *at least* k inversion pairs with threshold `min_threshold`. If no such integer exists, return -1 .

Input:

- The first line contains two integers n and k .
- The second line contains n integers: `nums`.

Output: Print a single integer — the minimum threshold, or -1 if impossible.

Example 1:

```
Input:
7 7
1 2 3 4 3 2 1

Output:
2
```

Explanation: For threshold $x = 2$, the valid inversion pairs are:

$$(3, 4), (2, 5), (3, 5), (4, 5), (1, 6), (2, 6), (4, 6), (5, 6).$$

There are less than $k = 7$ inversion pairs if we choose any integer less than 2 as the threshold.

Example 2:

```
Input:
5 4
10 9 9 9 1

Output:
8
```

Explanation: For threshold $x = 8$, the valid pairs are:

$$(0, 1), (0, 2), (0, 3), (1, 4), (2, 4), (3, 4).$$

There are less than $k = 4$ inversion pairs if we choose any integer less than 8 as the threshold.

Constraints:

- $1 \leq n \leq 10^4$
- $1 \leq \text{nums}[i] \leq 10^9$
- $1 \leq k \leq 10^9$